Signals and Systems

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$x[n-n_0]$	$z^{-n_0}X(z)$	$z \in R_x$ but may exclude $z = 0,\ \infty$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$

1. *This question caries 40% of the marks.*

a) Let $x(t)$ be a continuous-time signal. For each of the following cases of $x(t)$, state whether the signals are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. Give justifications for your statements. If periodic, state the period.

i) $x(t) = 2 + 2|\sin(\pi t)|$ [2]

Solution:
noncausal, periodic with period 1 s, even symmetric

ii) $x(t) = \begin{cases} e^{-3t} & t \geq 1 \\ 0 & t < 1 \end{cases}$ [2]

Solution:
causal, nonperiodic, nonsymmetric

iii) $x(t) = -2e^{t^2}$ [2]

Solution:
Non-causal, non-periodic, even symmetric

b) A continuous-time signal $x(t)$ is given by

$$x(t) = \begin{cases} 1-t & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find expressions for $y(t)$, given

i) $y(t) = x(-2t-1)$ [2]

Solution:

$$y(t) = x(-2t-1) = \begin{cases} 1-[-2t-1] = 2t+2 & \text{for } -1 \leq t \leq -0.5 \\ 0 & \text{otherwise} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow (t-1)$: shift in the direction of +ve time by 1
- $t \leftarrow (2t)$: compress on time axis by factor of 2
- $t \leftarrow (-t)$: reflection in $t = 0$

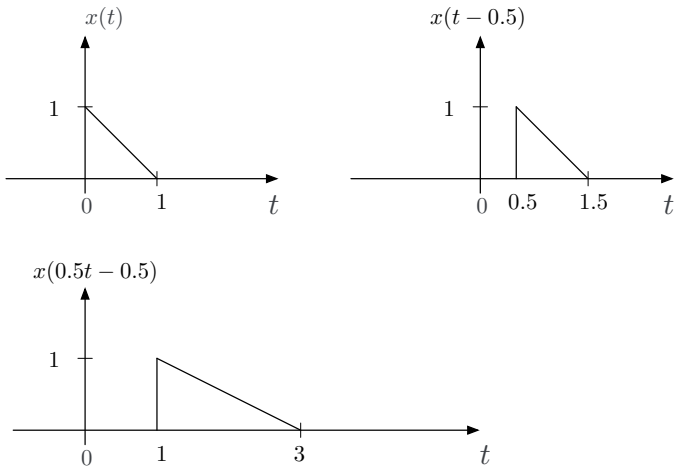
ii) $y(t) = x(0.5t - 0.5)$ [2]

Solution:

$$y(t) = x(0.5t - 0.5) = \begin{cases} 1 - [0.5t - 0.5] = -0.5t + 1.5 & \text{for } 1 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow (t - 0.5)$: shift in the direction of +ve time by 0.5
- $t \leftarrow (0.5t)$: expand on time axis by factor of 2



iii) For both question parts i) and ii), state the sequence of steps taken when transforming $x(t)$ to $y(t)$. [2]

c) The discrete-time signal $r[n]$ is defined to be

$$r[n] = \frac{\sin(\lambda n)}{\pi n}.$$

i) Write down a general expression for the energy of $r[n]$. [2]

ii) Let $R(e^{j\Omega})$ be defined as the DTFT of $r[n]$. Write down Parseval's theorem as it relates to $R(e^{j\Omega})$. [2]

iii) Use Parseval's theorem to find a closed-form expression for the energy of $r[n]$. Your solution should not contain any integrals or summations. [5]

Solution:
(i) Energy is given by

$$\sum_{n=-\infty}^{\infty} |r[n]|^2$$

(ii)

$$\sum_{n=-\infty}^{\infty} |r[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |R(e^{j\Omega})|^2 d\Omega$$

(iii) Energy is given by Parseval's relation using $R(e^{j\Omega})$ as

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |R(e^{j\Omega})|^2 d\Omega.$$

With $r[n]$ being a sinc function, it is known from the properties of the sinc function that

$$R(e^{j\Omega}) = \begin{cases} 1, & |\Omega| \leq \lambda \\ 0, & \lambda < |\Omega| \leq \pi. \end{cases}$$

Hence the energy is given by

$$\frac{1}{2\pi}\int_{-\lambda}^{\lambda} 1 d\Omega = \lambda/\pi.$$

d) Given that $u[n]$ represents the discrete-time unit step function and β is a constant, find the z-transform of

$$\cos(\beta n)u[n].$$

[5]

Solution:
Write $\cos(\beta n) = 0.5(e^{j\beta n} + e^{-j\beta n})$. Then use the standard z-transform relation that

$$\alpha^n u[n] \leftrightarrow \frac{1}{1-\alpha z^{-1}} \quad |z| > |\alpha|.$$

Set α to be $e^{j\beta}$ to obtain the z-transform

$$0.5\left(\frac{z}{z-e^{j\beta}} + \frac{z}{z-e^{-j\beta}}\right) = \frac{z(z-\cos(\beta))}{z^2 - 2z\cos(\beta)+1} \quad |z| > 1.$$

e) i) Draw a labelled sketch of the continuous-time signal

$$x(t) = -2tu(t).$$

[3]

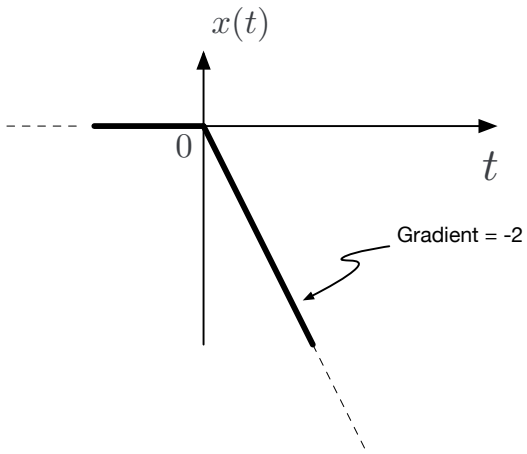
ii) Draw a labelled sketch of the continuous-time signal

$$x(t) = tu(t-2).$$

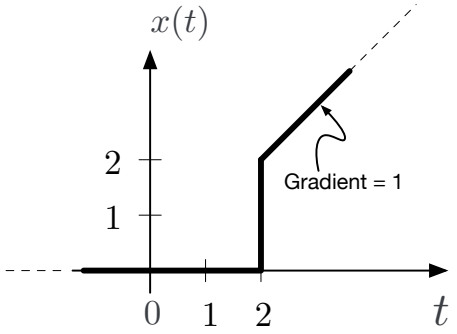
[3]

Solution:

(i)



(ii)



f) A person is talking in a room and the sound of their voice is sensed by a closely positioned microphone. The microphone signal is converted to a discrete-time signal $s[n]$ using an ADC, and stored. The resulting speech signal has a duration of 32 s.

i) Frame-based processing of the signal is proposed in which the signal is first segmented in time into frames by applying non-overlapping rectangular windows, and then each frame of the signal is processed or analyzed in sequence. Determine the window duration in seconds required to give a frame rate of 25 frames per second. [2]

ii) When the person was talking, they did not speak continuously but paused a few times to think. From the methods of analyzing signals studied in this course, briefly outline a method to classify each frame of the stored signal to be either a speech frame (during which the person was speaking) or a pause frame (during which the person was pausing). Give the mathematical expression for any analysis functions used in your method. [3]

iii) Now assume that the ADC uses a sampling frequency of 16 kHz. In a new frame-based analysis experiment, the frame duration was fixed at 24.375 ms and the STFT was computed for each frame of samples.

What is the frequency resolution of the resulting spectrum?

What is the lowest non-zero frequency that can be analyzed in this experiment?

What is the highest frequency that can be analyzed in this experiment.

Give your answers to 2 decimal places. [3]

Solution:
(i) 0.040 s

(ii) Find the energy per frame and compare it to a threshold. Energy in the frame is defined as

$$E = \sum_{-\infty}^{\infty} s^2[n] = \sum_{Frame} s^2[n].$$

2 marks for energy thresholding, +1 for reasonable enhancements such as band limiting the energy computation to the speech spectrum.

(iii) The frame duration of 24.375 ms corresponds to a frequency resolution of

$$\frac{1}{24.375 \times 10^{-}3} = 41.03 \text{ Hz}.$$

This is also the lowest frequency bin, so lowest frequency of analysis = 41.03 Hz. Highest frequency is determined by the Nyquist frequency of the ADC. Using an ideal anti-aliasing filter, the highest frequency is $8000 - 41.03 = 7958.7.00$ Hz. Full marks will be given also if non-ideal anti-aliasing filters are considered so that the highest frequency would be up to 15% lower.

2. *This question caries 30% of the marks.*

a) State and explain the differences between the Fourier transform and the Laplace transform. Include in your answer a fully labelled sketch of the s-plane and the relevant formulae with explanations. [6]

Solution:
The sketch should show the real σ and imaginary $j\omega$ axes in the s-plane and it should be clear the the Fourier transform concerns only the $\sigma = 0$ case. The key differences should include reference to, at least, the differences in the basis functions and the ability of Laplace representations to characterize both stable and unstable (dynamic) systems. The formulae for both transforms must give given and explained.

b) i) Define what is meant by a causal signal. [3]

ii) Let $\delta(t)$ be a continuous function of time for which the symbols have their usual meaning. Show that the Laplace transform of $\delta(t)$ is 1. [3]

iii) Consider the causal signal $x(t)$ for which the Laplace transform is known to be

$$X(s) = \frac{s+3}{s-3}.$$

Using the result of part ii), find the expression for $x(t)$ and state the Region of Convergence. [7]

Solution:
(i) Causal signals can be defined as zero for $t < 0$.
(ii)

$$\int_{-\infty}^{\infty} \delta(t) e^{-st} dt = 1 \times e^{-s0} = 1.$$

(iii)

$$X(s) = \frac{s+3}{s-3} = \frac{s-3+6}{s-3} = 1 + \frac{6}{s-3}$$
$$\Longrightarrow x(t) = \delta(t) + 6e^{3t}u(t)$$
$$\text{ROC} : \mathfrak{Re}(s) > 3.$$

c) Consider the signal $f(t)$ shown in Figure 2.1, and its Laplace transform denoted $F(s)$. Also consider the unit step function $u(t)$.

i) Draw labelled sketches of $\frac{df(t)}{dt}$ and $\frac{d^2f(t)}{dt^2}$. [4]

ii) Write an equation for $\frac{d^2f(t)}{dt^2}$ in terms of time-shifted versions of $\delta(t)$, where $\delta(t) = \frac{du(t)}{dt}$. [3]

iii) Hence, find an expression for the Laplace transform $F(s)$. State any properties of Laplace transforms used in obtaining this expression. *(Hint: consider taking Laplace transforms of both sides of your expression from part(ii)).* [4]

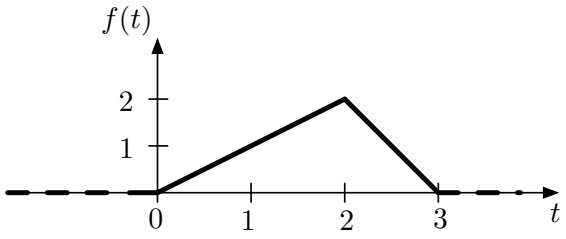
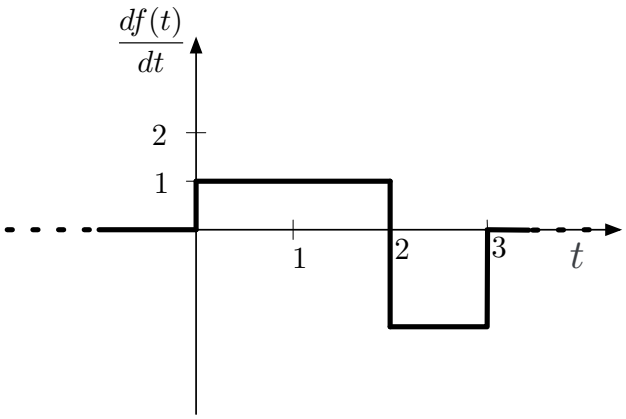
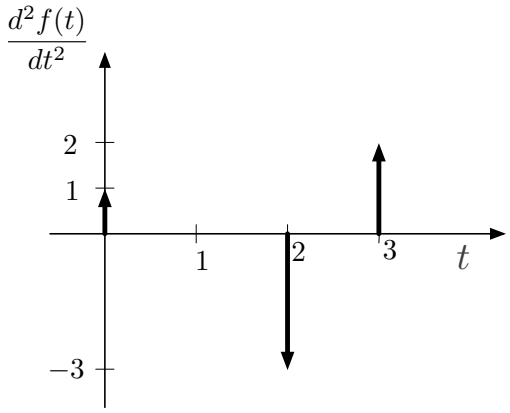


Figure 2.1

Solution:

(i)





(ii)

$$\frac{d^2 f}{dt^2} = \delta(t) - 3\delta(t-2) + 2\delta(t-3)$$

(iii)

$$F(s) = \frac{1}{s^2}(1 - 3e^{-2s} + 2e^{-3s}).$$

Properties used: differentiation and time-shift.

3. *This question caries 30% of the marks.*

a) In a clinical research project, it is proposed to study the flight patterns of harmful mosquitos by sensing and recording in discrete time the sound made by the mosquitos during flight in a defined volume.

i) Sketch and label a block diagram showing the main functions of the sound sensing and recording system that would be appropriate. [4]

ii) Explain how you would determine the sampling frequency for the system. [2]

iii) When testing the system, it is found to be unsatisfactory because:

- when the sound of the mosquitos is loud enough to be analysed, it is very badly corrupted by noise;
- when the noise is low enough not to be problematic, the sound of the mosquitos is too quiet to be analysed.

Explain what is causing this situation, and propose a solution. [4]

Solution:
(i) The sketch should show: one or more microphones in the defined volume, each with a microphone amplifier and a band-limiting analogue filter (anti-aliasing); an ADC with a given sampling frequency and number of output bits; a digital recorder (no details needed).
(ii) The sampling frequency can be determined when the bandwidth of the sound of mosquito flight is known so as to satisfy the Nyquist sampling criterion. A test recording at very high sampling frequency can be used to determine the relevant bandwidth. The sampling frequency can the be set to be greater than twice the highest frequency sound of the mosquitos.
(iii) Turning up the gain of the mic amplifier(s), also increases the noise in the measurement. This can be overcome by using microphones with lower self-noise and / or placing the microphones closer to the sound.

b) A discrete-time filter with system function $F(z)$ has an impulse response given by

$$f[n] = nu[n] - (n-6)u[n-6] - 6u[n-6].$$

i) Draw a labelled sketch of $f[n]$ over the range $n = 0, 1, 2, \ldots, 10$. [6]

ii) Starting with the z-transform of $u[n]$, use the differentiation property to determine the z-transform of $nu[n]$. Show your working. [5]

iii) Hence find an expression for $F(z)$. [5]

iv) Determine the gain of the filter at $\omega = 0$ and $\omega = \pi$. [4]

Solution:
(i) The values to be sketched for $f[n]$ are $[0,1,2,3,4,5,0,0,0,0,0]$. The sketch must be correctly labelled for full marks.
(ii) Starting with

$$u[n] \leftrightarrow \frac{z}{z-1}$$

and using the property

$$nx[n] \leftrightarrow -z\frac{d}{dz}X(z),$$

differentiation of the quotient leads to

$$nu[n] \leftrightarrow -z.\frac{-1}{(z-1)^2} = \frac{z}{(z-1)^2}.$$

(iii) Using linearity, the z-transform can be found as the sum of the z-transform of each term:

$$\begin{aligned} nu[n] &\leftrightarrow \frac{z}{(z-1)^2} \\ (n-6)u[n-6] &\leftrightarrow z^{-6}\frac{z}{(z-1)^2} = \frac{1}{z^5(z-1)^2} \\ 6u[n-6] &\leftrightarrow \frac{6}{z^5(z-1)}. \end{aligned}$$

Hence

$$F(z) = \frac{z^6 - 6z + 5}{z^5(z-1)^2}.$$

(iv) Frequencies $\omega = 0, \pi$ correspond to $z = +1, -1$. In order to solve this problem, we have to handle term $(z-1)$ specially as it will approach zero as z approaches 1. The numerator can be factored to give

$$F(z) = \frac{(z-1)^2(z^4+2z^3+3z^2+4z+5)}{z^5(z-1)^2}.$$

so that the $(z-1)^2$ terms cancel in the numerator and denominator. This factorization is hinted at by the $(z-1)$ term in the denominator. Then

$$\begin{aligned} |F(1)| &= \frac{1+2+3+4+5}{1} = 15 \text{ or } 23.5 \text{ dB and} \\ |F(-1)| &= \frac{1-2+3-4+5}{|-1|} = 3 \text{ or } 9.5 \text{ dB}. \end{aligned}$$